

Fixed-Ro equal-area theory on the equatorial β plane, for the SS09 shallow-water model

pengcheng-ss09 suite, Tier 0 (A0 memo)

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Beta-plane translations of the closed forms in the fixed-Ro manuscript (`papers/fixed-ro-eq-area_ms-draft-v1.pdf`, “ms” below), derived against this model’s exact equations. Every result carries a basis tag; “pending run” tags get upgraded as the suite’s runs confirm them. Section 8 lists the falsifying checks and their status, and the run record follows it.

1 Setup and notation

Steady state, hemispherically symmetric forcing ($y_0 = 0$), northern hemisphere ($y > 0$). The model’s steady equations (`SCIENCE.md` section 2), with $T = \theta/1.6$ and the v equation’s factor of two multiplying only the time tendency (so it drops out of every steady balance):

$$v \left(\beta y - \frac{\partial u}{\partial y} \right) = \mathcal{H}(\theta_E - \theta) u \frac{\partial v}{\partial y} + \epsilon_u u + v_d \mathcal{H}(u) \operatorname{sgn}(y) \frac{\partial u}{\partial y}, \quad (1)$$

$$\beta y u = -\frac{gH}{T_0} \frac{\partial T}{\partial y} + k_v \frac{\partial^2 v}{\partial y^2}, \quad (2)$$

$$\frac{\delta \Delta_z}{H} \frac{\partial v}{\partial y} = \frac{\theta_E - \theta}{\tau}. \quad (3)$$

Forcing (SS09 parabolic profile, the theory-matched choice): $\theta_E = \theta_{00} - \Delta_y (y/y_1)^2$ for $|y| < y_1$. In temperature units,

$$T_E(y) = T_E(0) - b y^2, \quad b \equiv \frac{\Delta_T}{y_1^2}, \quad \Delta_T \equiv \frac{\Delta_y}{1.6}, \quad (4)$$

with $T_0 = 300$ K. The small-angle limit of the ms is exact here: the β plane has no $\cos \varphi$ factors, so no truncation enters any of the following.

2 The one-line mapping

Every ms result carries over with the substitutions

$$\operatorname{Ro}_{\text{th}} \rightarrow R_\beta \equiv \frac{4gH\Delta_T}{T_0\beta^2 y_1^4}, \quad \Delta_h \theta_0 \rightarrow \Delta_T, \quad a\varphi \rightarrow y, \quad \varphi_{\text{Ro}} \rightarrow Y_{\text{Ro}}/y_1. \quad (5)$$

[**derived**] R_β follows from the sphere definition $\operatorname{Ro}_{\text{th}} = gH\Delta_h/(\Omega a)^2$ with $\Omega = \beta a/2$ and $\Delta_h = \Delta_T a^2/(T_0 y_1^2)$ (matching the forcing curvature at the equator); the planetary radius cancels, as it must on a β plane. At this repo’s defaults ($g = 9.81 \text{ m s}^{-2}$, $H = 16 \text{ km}$, $\Delta_y = 50 \text{ K}$ so $\Delta_T = 31.25 \text{ K}$, $T_0 = 300 \text{ K}$, $\beta = 2 \times 10^{-11} \text{ m}^{-1} \text{ s}^{-1}$, $y_1 = 9439 \text{ km}$):

$R_\beta = 0.0206.$

(6)

3 Wind and temperature

[derived: pending run confirmation] Uniform $\text{Ro} = (\partial_y u)/(\beta y)$ integrates to $u_{\text{Ro}}(y) = \text{Ro} \beta y^2/2$ (ascent at $y = 0$). Gradient balance (neglecting k_v) gives $\partial_y T = -(T_0/gH) \text{Ro} \beta^2 y^3/2$, so

$$T_{\text{Ro}}(y) = T_{\text{Ro}}(0) - c y^4, \quad c \equiv \frac{\text{Ro} T_0 \beta^2}{8gH}. \quad (7)$$

4 Equal-area edge and equatorial temperature

[derived: confirmed by runs 1a–1b at the equilibrated day-6000 record: depression within 4%, edge -7 to -8% ; at profile level the inner half of the cell agrees at the 10% level and the terminus misfit is ~ 32 – 36% of the depression; run-record addenda] With $D(y) = T_{\text{Ro}}(y) - T_E(y)$, imposing $D(Y) = 0$ and $\int_0^Y D dy = 0$ (measure dy ; the Newtonian relaxation has uniform τ) gives $\frac{2}{3}bY^2 = \frac{4}{5}cY^4$, i.e. $Y^2 = \frac{5}{6}b/c$, which in Section-2 variables is

$$Y_{\text{Ro}} = y_1 \left(\frac{5R_\beta}{3\text{Ro}} \right)^{1/2} \quad [\text{ms Eq. 13 analog}], \quad (8)$$

$$T_E(0) - T_{\text{Ro}}(0) = \frac{5}{18} \frac{R_\beta \Delta_T}{\text{Ro}} \quad [\text{ms Eq. 15 analog}]. \quad (9)$$

Defaults, $\text{Ro} = 1$: $Y_{\text{Ro}} = 1.75 \times 10^6$ m and equatorial depression 0.18 K (in T ; multiply by 1.6 for θ units). At $\text{Ro} = 0.15$ the depression is 1.2 K (T) / 1.9 K (θ).

Consistency note: the analogous derivation for the $\sin^2$ profile needs the numerical equal-area solver (puffins `equal_area_bouss` with Ro -scaled `del_h_over_ro`, verified in puffins PR #46). Near the equator $\sin^2(\pi y/(2y_1)) = (\pi y/(2y_1))^2 + O(y^4)$, so $\sin^2$ behaves like a parabola with curvature enhanced by $(\pi/2)^2 = 2.47$; Section 7's postdiction check uses exactly this factor.

5 Meridional wind, heat flux, momentum flux

[derived: confirmed by runs 1a–1b at the equilibrated day-6000 record: v_{max} within 2–7% and its location within 4–5%; at profile level the inner flank agrees at the 3–8% level and the terminus flank misfit is ~ 44 – 45% of v_{max} ; run-record addenda] The steady thermodynamic balance (3) integrates to the model's v profile directly, using $\theta_E - \theta = 1.6 (T_E - T) = -1.6 D$:

$$v(y) = -\frac{1.6H}{\delta \Delta_z \tau} \int_0^y D dy' = \frac{1.6H}{\delta \Delta_z \tau} \frac{5}{18} \frac{R_\beta \Delta_T}{\text{Ro}} Y_{\text{Ro}} (x - 2x^3 + x^5), \quad x \equiv \frac{y}{Y_{\text{Ro}}}. \quad (10)$$

The bracket is exactly the ms Eq. (16) shape; it peaks at $x = 5^{-1/2} = 0.447$ with value 0.286, and $v_{\text{max}} \propto \text{Ro}^{-3/2}$ (ms Eq. 20 analog). Defaults, $\text{Ro} = 1$: $v_{\text{max}} = 3.0 \times 10^{-3} \text{ ms}^{-1}$ at $y = 0.78 \times 10^6$ m.

Column fluxes, using the two-layer reading (upper slab of depth δ with velocity $+v$ and potential temperature θ_{top} , return slab of depth δ with $-v$ and θ_{bot} , $\theta_{\text{top}} - \theta_{\text{bot}} = \Delta_z$; return-flow u negligible):

$$\text{heat flux: } \int v \theta dz = \delta \Delta_z v(y) \quad [\text{ms Eq. 16}], \quad (11)$$

$$\text{momentum flux: } \int v u dz = \delta v(y) u_{\text{Ro}}(y) \quad [\text{ms Eq. 17}], \quad (12)$$

$$\text{vertical velocity: } w = \delta \frac{\partial v}{\partial y}, \quad w_{\text{max}} \propto \text{Ro}^{-1} \quad [\text{ms sec. 3f}]. \quad (13)$$

The ms’s counterintuitive momentum-flux result carries over: expanding $\delta v u_{\text{Ro}}$, the leading term in y is Ro-independent (the Ro^{-1} strengthening of v cancels the Ro weakening of u in the product) and $\partial(\text{flux})/\partial\text{Ro} < 0$ strictly inside the cell, exactly as verified for the sphere forms in puffs in `test_eq_area.py`.

6 Diagnosed quantities: conventions for the suite

- **Full-profile scorecard (the confirmation standard for every suite run):** the simulated u , T , and v profiles are compared against the zero-free-parameter theory profiles at the diagnosed (regression) cell-mean Ro, reporting $\max|\Delta|$ and its location both over the cell $[0, Y_{\text{Ro}}]$ and over its inner half, each scaled by the field’s theory amplitude (cell-edge u , the equatorial depression, and v_{max} respectively). The temperature comparison is visualized as the equal-area lobes $D(y) = T - T_E$, which removes the ~ 200 K shared background. Implemented in `scripts/fixed_ro_scorecard.py` (one four-panel figure per run; fields are means over the final 30 days). Scalar anchors (depression, v_{max} , edge) are by-products of the same tool, not the confirmation standard: runs 1a–1b passed the scalars at the 1–8% level while the profile misfit at the terminus is ~ 35 –45% of scale (see the profile addendum in the run record).
- **Run duration (budget-grade):** the steady-state detector’s KE and T -variance metrics are nearly blind to the slow jet-position mode: in runs 1a–1b the detector fired at day ~ 960 with the jet latitude still moving $\sim 0.7\%$ per 60 d (measured over the final 120 d). Scorecard-grade runs therefore use a fixed length of at least five times the slowest damping timescale, the drag time $1/\epsilon_u$ (1157 d at the default $\epsilon_u = 10^{-8} \text{ s}^{-1}$, so 6000 d), with no early stop. The scorecard enforces this after the fact: it fits the jet-latitude trend over the final 120 d and warns unless the implied remaining drift (trend $\times 1/\epsilon_u$, the exponential-tail estimate) is below 0.2%. The detector remains fine for smokes and exploratory runs.
- **Local Ro(y):** $(\partial_y u)/(\beta y)$, the model’s existing diagnostic (NaN within $|y| < \Delta y_{\text{grid}}$ of the equator).
- **Cell-mean Ro, two definitions, both reported, one fixed interval:** both are computed over $[y_5, y_{\text{jet}}]$ with $y_5 \equiv a \times 5^\circ = 5.56 \times 10^5 \text{ m}$, the beta-plane analog of Hill et al. (2025)’s convention (average from the descending edge to whichever of the ascending edge or 5° is closer, Ro being ill-defined near the equator; for the symmetric cells here the 5° cutoff binds). (i) The area mean of local Ro; (ii) the best-fit Ro regressing $u(y)$ against $\beta y^2/2$, the *primary* definition (insensitive to the interval). The earlier inner anchor $y_{\text{Ro-max}}$ (argmax of the Ro profile) was retired 2026-07-17: an argmax is a discontinuous functional of a smooth profile, and it jumped $1.1 \times 10^6 \text{ m}$ between epochs of run 1a when the profile’s two gentle extrema traded rank, moving the area mean $0.476 \rightarrow 0.355$. On the fixed interval the two equilibrated runs’ area means nearly coincide (0.429, 0.424).
- **Cell edge, three definitions, all reported:** the v -threshold edge ($|v|$ below 10% of its extremum, existing diagnostic; see run 1a for its failure mode at small v_d), the jet latitude (existing spline diagnostic), and the θ -merge latitude (where θ rejoins θ_E within a tolerance, searched poleward of the *jet*; see run 1a). The theory says they coincide; their splitting is itself a result.

- **Momentum-budget shares:** each steady term of (1) evaluated from output; the drag share $\epsilon_u u$ and the vertical-advection share quantify the accuracy of the EMFD–Coriolis balance underlying Zhang et al. (2025) Eq. 24, $\text{Ro} = v/(v + v_d)$.

7 Emergent cell-mean Ro closure

[derived: confirmed by the Tier-1 v_d ladder, 2026-07-20 run record: within 4% of the diagnosed $\bar{\text{Ro}}$ for $v_d \geq 0.05$, and within 5% at every rung once the drag extension below is included] Combining Zhang’s Eq. 24 at the cell-mean level with Section 5’s v magnitude (the cell mean of the bracket over $[0, 1]$ is $1/6$):

$$\bar{\text{Ro}} = \frac{\bar{v}}{\bar{v} + v_d}, \quad \bar{v}(\bar{\text{Ro}}) = \frac{1.6H}{\delta\Delta_z\tau} \frac{5}{18} \frac{R_\beta\Delta_T}{\bar{\text{Ro}}} \frac{Y_{\text{Ro}}(\bar{\text{Ro}})}{6}, \quad (14)$$

a transcendental equation for $\bar{\text{Ro}}(v_d)$ with no simulation input. This is the suite’s analysis G; it extends Zhang et al. (2025) section 4 (their max-Ro scalings $v_d^{-2/5}$, $v_d^{-1/4}$) to the cell-mean variable the ms uses. Because run 1b showed the default drag to be leading-order at this forcing, Tier 1 also evaluates a drag-extended variant [derived: cell-midpoint drag]: the steady balance with the drag term retained and $u = \text{Ro} \beta y^2/2$ gives the pointwise relation $\text{Ro}(y) = v/(v + v_d + \epsilon_u y/2)$, whose cell-mean analog with $y \rightarrow Y_{\text{Ro}}/2$ is $\bar{\text{Ro}} = \bar{v}/(\bar{v} + v_d + \epsilon_u Y_{\text{Ro}}/4)$.

Postdiction check [computed]: Section 5’s v with the $\sin^2$ profile’s larger near-equator curvature ($(\pi/2)^2 = 2.47$ in b , entering v through $b^2 Y/c$ as a factor ~ 9.6) predicts $v_{\text{max}} \sim 0.03 \text{ m s}^{-1}$ for the published defaults, matching Zhang et al. (2025) Figs. 2–3 magnitudes within $\sim 1.5\times$.

8 Falsifying checks and status

Every run in the checks below also gets the Section 6 full-profile scorecard on u , T , and v ; the tier-specific items are the additional, targeted comparisons.

1. **Full $u/T/v$ profiles, edge, and equatorial depression vs theory at diagnosed Ro (the most-falsifying single check):** SS09-parabolic forcing, $v_d = 0$. Framing correction from the first run: with the default vertical momentum advection on, $v_d = 0$ is *not* the $\text{Ro} = 1$ limit (Zhang et al. 2025: $u \sim \beta y^2/3$ near the ascent, and drag lowers it further); the theory must be evaluated at the run’s diagnosed cell-mean Ro. The $\text{Ro} \rightarrow 1$ anchor requires `--no-vert-advect-u` and reduced drag (run 1b). Status: runs 1a–1b below, scalar level and profile level (addendum).
2. **Edge collapse:** $Y_{\text{edge}}(\text{Ro}/R_\beta)^{1/2}/y_1 = (5/3)^{1/2}$ flat across the v_d , Δ_y , β , and radius ladders. Status: v_d arm done (Tier 1, 2026-07-20): the jet-based collapse constant is flat to 2.6% across $\bar{\text{Ro}} = 0.054\text{--}0.480$ but sits 6–8% below $(5/3)^{1/2}$ (1.19–1.22 vs 1.291), i.e. the scaling holds with a systematic offset. Δ_y , β , radius arms pending Tier 2.
3. **Radius-ladder null:** dimensional Y_{edge} invariant under $(\beta \rightarrow \beta/k, y_1 \rightarrow ky_1, L \rightarrow kL)$. Status: pending Tier 2.
4. **v -profile shape and $\text{Ro}^{-3/2}$ magnitude scaling** (Section 5). Status: shape checked at profile level in runs 1a–1b (addendum); scaling confirmed by the Tier-1 ladder with a growing deficit (2026-07-20): effective exponent -1.40 over $\bar{\text{Ro}} = 0.054\text{--}0.480$, with the zero-free-parameter prediction overpredicting v_{max} by 2% at $v_d = 0.0125$ growing to 14% at $v_d = 2.5$.

5. **Momentum flux increases as Ro decreases** (Section 5). Status: confirmed (Tier 1, 2026-07-20): peak $\delta v u$ rises monotonically from 511 to 8692 m³ s⁻² (17 $\times$) as $\bar{\text{Ro}}$ falls 0.480 $\rightarrow$ 0.054, with the theory at fitted $\bar{\text{Ro}}$ within 8% for $v_d \geq 0.025$.
6. **Emergent $\bar{\text{Ro}}(v_d)$** (Section 7). Status: confirmed (Tier 1, 2026-07-20): the drag-free closure is within 4% of the diagnosed $\bar{\text{Ro}}$ for $v_d \geq 0.05$ (and +11% at $v_d = 0.0125$); the drag-extended variant is within 5% at every rung.

Run record

Each confirming or refuting run gets an entry: configuration, output path, measured vs predicted.

2026-07-16, check 1a: $v_d = 0$, default physics (vertical advection on)

Config: SS09 parabolic forcing, $v_d = 0$, $n_y = 801$, $\Delta t = 30$ s, numba backend, steady-state stop (window 30 d, threshold 10^{-4}), converged day 959. Output: `model_output/fixed_ro_suite/tier0_amc_edge/amc_ss09_ny801_dt30_vd0.nc`. Steadiness caveat [**computed**]: the jet was still drifting +0.4% per 60 d at stop (drag-timescale tail), so numbers carry roughly a 2% allowance.

Diagnosed: cell-mean Ro 0.469 (area mean) / 0.470 (u regression); local Ro(y) is *not* uniform: ~ 0.68 at the ascent (the vertical-advection 2/3 of Zhang et al. 2025, drag-reduced), declining through ~ 0.6 mid-cell to ~ 0.2 approaching the jet, i.e. the ms’s linear-Ro motivation in miniature.

Theory at diagnosed Ro = 0.470 vs run [**computed**]:

quantity	predicted	measured	rel. gap
equatorial depression $T_E(0) - T(0)$	0.381 K	0.379 K	0.5%
$v_{\max}$	9.28×10^{-3} m/s	9.39×10^{-3} m/s	1.2%
y of $v_{\max}$ ($0.447 Y_{\text{Ro}}$)	1.14×10^6 m	1.18×10^6 m	3.5%
cell edge Y_{Ro} (vs jet latitude)	2.55×10^6 m	2.36×10^6 m	-7.4%

Sections 4–5 upgraded to [**confirmed at the 1–8% level, single run**] for these observables at this Ro. Additional findings: the u plateau outside the cell is the parabolic forcing’s RCE gradient wind,

$$u_{\text{RCE}} = \frac{2gH\Delta_T}{T_0\beta y_1^2} = 18.35 \text{ m/s}, \quad (15)$$

constant out to y_1 (verified to 3 digits); the v -threshold edge diagnostic is unusable at small v_d because the drag circulation $v = \epsilon_u u_{\text{RCE}}/(\beta y) \sim 2 \times 10^{-3}$ m/s exceeds 10% of $v_{\max}$ out to $\sim 9.3 \times 10^6$ m; the θ -merge diagnostic must search poleward of the jet, not of the v max, to avoid the equal-area lobe crossing.

2026-07-16, check 1b: $v_d = 0$, vertical advection off

Config: as 1a plus `--no-vert-advec-u`; converged day 971. Output: `.../tier0_amc_edge/amc_ss09_ny801_dt30_vd0_novert.nc`.

Removing vertical advection left the cell-mean Ro essentially unchanged (fit 0.473 vs 0.470; area mean 0.536 vs 0.469), because **the Rayleigh drag, at its default $\epsilon_u = 10^{-8} \text{ s}^{-1}$, is the**

leading-order non-AMC agent for this forcing [computed]: the steady zonal momentum balance over the cell interior is

$$v \left(\beta y - \frac{\partial u}{\partial y} \right) = \epsilon_u u \quad \text{to within } \sim 5\text{--}10\% \quad (16)$$

(median ratio 1.05 over $0.3 \times 10^6 < y < 2.2 \times 10^6$ m). The parabolic forcing’s weak circulation ($v_{\max} \sim 9 \times 10^{-3}$ m/s) makes $(1 - \text{Ro}) \sim \epsilon_u u / (v \beta y)$ order one. Local $\text{Ro}(y)$ declines monotonically from ~ 1 at the ascent to ~ 0.2 at the jet.

Theory at diagnosed $\text{Ro} = 0.473$ vs run **[computed]**: depression 0.378 predicted vs 0.358 K measured (5.3%); $v_{\max}$ 9.18×10^{-3} vs 8.81×10^{-3} m/s (4.2%); edge 2.54×10^6 vs jet 2.36×10^6 m (−7.1%).

Consequences for the suite: (i) the $\text{Ro} \rightarrow 1$ anchor needs reduced ϵ_u , so the Tier-3 drag ladder is load-bearing rather than optional; (ii) at the $\sin^2$ forcing the circulation is $\sim 3\times$ stronger, so the drag share is $\sim 3\times$ smaller but still non-negligible at $v_d = 0$ (consistent with Zhang et al. 2025 Fig. 4’s max $\text{Ro} \sim 0.9$ there); (iii) the steady-state detector (window 30 d, 10^{-4}) fires while the jet still drifts $\sim 0.4\%$ per 60 d on the drag timescale, so budget-grade runs need a stricter threshold or a fixed post-detection extension.

2026-07-16, profile-level addendum to runs 1a–1b

The tables above are endpoint checks (the equatorial depression and $v_{\max}$ are, respectively, the $y = 0$ anchor of T and a single point of v). The suite standard is now the full-profile scorecard (Section 6), applied retroactively to both runs: final-30-day means, theory at the fitted cell-mean Ro of 0.479 (1a) and 0.473 (1b). Under this standardized window 1b’s fit reproduces the value quoted above exactly, while 1a’s shifts from 0.470 to 0.479, within run 1a’s stated $\sim 2\%$ drift allowance. Profile results **[computed]**:

run	field	max $ \Delta $ (cell)	at y [10^6 m]	% of scale	inner half
1a	u	10.4 m/s	2.52	34	2.3 m/s (7.5%)
1a	T	0.133 K	2.01	36	0.038 K (10%)
1a	v	4.0 mm/s	2.44	45	0.49 mm/s (5.5%)
1b	u	10.2 m/s	2.52	33	3.2 m/s (10%)
1b	T	0.152 K	2.01	40	0.042 K (11%)
1b	v	3.9 mm/s	2.44	43	0.48 mm/s (5.2%)

Scales: cell-edge $u_{\text{Ro}}(Y_{\text{Ro}})$, the equatorial depression, and $v_{\max}$. Reading: the inner half of the cell agrees at the 5–11% level in all three fields, and the misfit concentrates at the terminus in all three, consistent with the diagnosed local $\text{Ro}(y)$ declining to ~ 0.2 at the jet: a single cell-mean Ro is furthest from the local value exactly there. The simulated subtropical warm lobe of $D(y)$ peaks at 0.17 K (1a) against the predicted 0.30 K and is smeared poleward across the edge (D stays $\sim +0.015$ K out to at least 4.5×10^6 m), suggesting the sharp-edge equal-area closure, not the quartic interior shape, is the limiting idealization. The u misfit sits at the theory’s terminus discontinuity (the jump to the RCE plateau). The ms’s linear- Ro variant targets exactly this regime and is testable against these same stored profiles (suite extension, not yet scheduled).

2026-07-17, drift fix: runs 1a–1b extended to day 6000

The detector-stop states above were drift-contaminated: at the day ~ 960 stop the jet latitude was still moving -0.69% per 60 d (fit over the final 120 d; equatorward), which the exponential-

tail estimate turns into $\sim 13\%$ remaining drift. (This measurement supersedes the “+0.4% per 60 d” figure quoted in the 1a entry, which came from a different ad-hoc check.) Both runs were therefore restarted from their day-959/971 checkpoints and integrated to day 6000, about five drag timescales, with a fixed length and no early stop, per the Section 6 duration policy. The `..._day6000.nc` outputs are now the citable record for checks 1a–1b; both pass the drift gate with a flat jet trend (+0.0000% per 60 d) and the fitted Ro stable to $< 0.001\%$ between consecutive 30-d windows.

What the contamination had been hiding **[computed]** (numbers restated 2026-07-17 under the fixed $[y_5, y_{\text{jet}}]$ interval of Section 6): the jet settled equatorward (1a: $2.36 \rightarrow 2.33$; 1b: $2.38 \rightarrow 2.33 \times 10^6$ m), so the edge gap vs Y_{Ro} *widens* to -7.7% (1a) / -7.2% (1b); 1b’s fitted Ro rises $0.473 \rightarrow 0.485$ (+2.5%, the largest single shift; 1a’s is 0.480); the depression gaps become +3.5% (1a: measured 0.385 vs predicted 0.372 K) and -0.8% (1b: 0.366 vs 0.369 K); and v_{max} gaps become 6.8% (1a: 9.58 vs 8.97 mm/s) and 2.0% (1b: 9.03 vs 8.85 mm/s). Equilibrated profile metrics, superseding the table above:

run	field	max $ \Delta $ (cell)	at y [10^6 m]	% of scale	inner half
1a	u	10.2 m/s	2.52	33	1.8 m/s (6.0%)
1a	T	0.121 K	2.01	32	0.037 K (9.8%)
1a	v	3.9 mm/s	2.44	44	0.72 mm/s (8.0%)
1b	u	9.6 m/s	2.48	31	2.5 m/s (8.1%)
1b	T	0.133 K	1.97	36	0.039 K (10.5%)
1b	v	3.9 mm/s	2.40	45	0.25 mm/s (2.9%)

The profile-level story is unchanged in character: inner half at the 3–11% level, terminus at 31–45% of scale, in all three fields of both runs. During the slow tail, 1a’s local-Ro profile also exposed a fragility in the then-current area-mean interval anchor; that finding and the resulting convention change are in the next entry.

2026-07-17, settling analysis, slow-drift gate, and cell-mean-Ro convention update

Settling [computed]: every quoted quantity (jet latitude, fitted Ro, depression, v_{max}) is within 0.1% of its final value by day ~ 2400 –2600 (1a) / 2700–3200 (1b), and within 0.02% by day ~ 3000 –3550 / 3450–3970, so the fixed 6000-d length is $\sim 2\times$ conservative. The tail is an underdamped oscillation: envelope timescale 394 d (1a) / 436 d (1b), about a third of the drag time, with the jet overshooting its final latitude (deviation lobes near days 1600, 3200, 4500 in 1a). The stop-time “ $\sim 13\%$ remaining” drag-timescale estimate above was therefore conservative by $\sim 3\times$; the realized post-stop movement was 1–3% per quantity. Reproduce with `scripts/tier0_settling_analysis.py`.

Slow-drift gate: a trend-based stopping criterion fails on this tail: replayed on run 1a it fires at day 1629, the crest of the first overshoot lobe, where the 120-d trend is ~ 0 while the jet is still 1.4% from its final position. The detector therefore gained an opt-in range criterion (`--slow-drift-gate`): converged only when jet latitude, $\max|v|$, and the equatorial depression each have relative range $< 0.2\%$ over the trailing $1/\epsilon_u$ (1158 d at default drag). Exact replay on the full 1a/1b histories (`scripts/tier0_gate_replay.py`): fires at day 3683 (1a; jet binding) / 3912 (1b; v_{max} binding), saving 39%/35% vs the fixed 6000 d with every gated quantity settled below 0.02%. Confirmed by a gated cold-start run of the 1a configuration (`..._gatecheck.nc`): it stopped at day 3683 exactly (the gated trajectory is bitwise-identical to the recorded one), and

its scorecard matches the day-6000 record to display precision in every reported quantity, with the drift gate passing at 0.06% implied remaining.

Convention update: the cell-mean-Ro interval anchor $y_{\text{Ro-max}}$ was retired for the fixed $[y_5, y_{\text{jet}}]$ interval (Section 6): 1a’s smooth $\text{Ro}(y)$ has a near-equator spike (1.00 at day 960, relaxing to 0.62) and a mid-cell bump ($0.65 \rightarrow 0.64$ at $y \approx 1.15 \times 10^6$ m), and when they traded rank the argmax anchor jumped 1.1×10^6 m, moving the area mean $0.476 \rightarrow 0.355$ from a 0.03 relative-height flip. On the fixed interval: area means 0.429 (1a) / 0.424 (1b), fitted 0.480 / 0.485.

2026-07-20, Tier 1: the v_d ladder (checks 2, 4, 5, 6)

Config: as run 1a (SS09 parabolic, $n_y = 801$, $\Delta t = 30$ s, numba, production formulation) with $v_d \in \{0.0125, 0.025, 0.05, 0.125, 0.25, 0.5, 1.25, 2.5\}$ m/s, Zhang et al. (2025)’s exact Fig. 4 ladder; run 1a’s day-6000 record serves as the $v_d = 0$ rung. Stopping: slow-drift gate (detector window 30 d, threshold 10^{-4} ; gate defaults), the policy validated by the 1a gate check. Gates fired at days 3138/2494/1944/1936/1926/1912/1875/1822 in ascending v_d ; the eight runs took 6.1 min wall on a three-worker numba pool (0.06–0.08 s/day contended). Every run passes the scorecard drift gate with a flat jet trend, holds $\max_y |u(y) - u(-y)| = 0$ exactly, and its 10-day smoke ran first. Whole-domain settling: $\max_y |\Delta u| \leq 1.4 \times 10^{-3}$ m/s between the final two 30-d windows, largest at the far-field notch described below. Outputs, figures (t1_ro_ladder.png, t1_closure.png, t1_scalings.png, t1_budget.png, t1_farfield.png), per-run scorecards, and t1_summary.json live in model_output/fixed_ro_suite/t1_vd_ladder/; reproduce with scripts/fixed_ro_t1_analysis.py.

Ladder [computed], all quantities final-30-day means:

v_d [m/s]	$\bar{\text{Ro}}$ fit	max Ro (in cell)	y_{jet} [10^6 m]	v_{max} [mm/s]	v_{max} pred [mm/s]	depr. [K]	depr. pred [K]	EMFD:drag
0	0.480	0.639	2.33	9.6	9.0	0.385	0.372	0.00:0.95
0.0125	0.338	0.450	2.83	14.9	15.2	0.524	0.530	0.49:0.49
0.025	0.279	0.379	3.12	18.9	20.3	0.614	0.642	0.63:0.35
0.05	0.226	0.309	3.47	25.2	27.8	0.746	0.792	0.75:0.23
0.125	0.166	0.228	4.06	39.1	44.3	1.000	1.079	0.87:0.12
0.25	0.129	0.179	4.61	56.1	64.5	1.274	1.387	0.92:0.07
0.5	0.100	0.139	5.25	81.9	94.8	1.640	1.793	0.95:0.04
1.25	0.071	0.098	6.20	137.1	158.5	2.312	2.525	0.98:0.02
2.5	0.054	0.075	7.01	203.6	236.4	3.011	3.297	0.98:0.01

Momentum-budget regimes [computed]: the band-mean shares of $v(\beta y - u_y)$ over $[y_5, y_{\text{jet}}]$ put the drag–EMFD crossover at $v_d = 0.0125$ exactly (shares 0.49:0.49); vertical advection contributes $\leq 5\%$ at every rung. So Zhang Eq. 24’s neglect of vertical advection is excellent throughout, and its neglect of drag fails only for $v_d \lesssim 0.05$. At this forcing the ladder never enters Zhang’s nonlinear AMC regime: as $v_d \rightarrow 0$ the drag takes over from the EMFD and $\bar{\text{Ro}}$ saturates at the drag-limited 0.480 (max local Ro 0.639) rather than approaching 1. Zhang’s $\sin^2$ ladder reaches max Ro ≈ 0.9 because that circulation is $\sim 3\times$ stronger, shrinking the drag share accordingly (run 1b consequence (ii)).

Max-Ro slopes vs the pre-registered targets [computed]: log-log fits over the four smallest / four largest v_d rungs give -0.297 / -0.377 for the in-cell max local Ro (-0.309 / -0.376 for the fitted cell-mean $\bar{\text{Ro}}$). The large- v_d branch agrees with Zhang’s eddy-dominant scaling ($v_d^{-2/5}$; their $\sin^2$ fit -0.41). The small- v_d branch is steeper than their AMC-branch -0.26 ($v_d^{-1/4}$), consistent with the budget: here the small- v_d arrest of Ro comes from drag, and the AMC branch their $-1/4$

law describes is not accessed at default ϵ_u . Testing the $-1/4$ law needs the Tier-3 reduced-drag ladder.

Geometry [computed]: the max-Ro latitude y_m coincides with the v -max latitude to within one grid point at seven of nine rungs (two at the rest), confirming Zhang’s identification of the two; $y_{\text{jet}}/(2y_m) = 1.02\text{--}1.11$ across the ladder, so their $y_E = 2y_m$ closure holds to $\sim 10\%$. The memo’s own v -peak prediction $0.447 Y_{\text{Ro}}$ matches the measured v -max latitude to 2–4% at every rung.

Closure (analysis G) [computed]: diagnosed $\bar{\text{Ro}}$ vs the no-simulation-input solutions of Eq. (14): G gives $+11.4/+7.3/+3.8/+1.4/+0.6/-0.1/-1.2/-1.6\%$ across the eight rungs (ascending v_d); the drag-extended variant gives $-4.9/-3.0/-2.3/-1.7/-1.2/-1.2/-1.7/-1.9\%$. The drag-free overshoot at small v_d is the neglected drag; the extension removes most of it. In its $v_d \rightarrow 0$ limit the extension predicts $\bar{\text{Ro}} \approx 0.40$ vs the measured 0.480, so the cell-midpoint drag weighting is $\sim 20\%$ too strong in the drag-dominated limit.

Profile level [computed]: the u misfit stays terminus-locked at 28–31% of the cell-edge scale at every rung while the inner half sits at 4.6–6.0%; the v misfit falls from 44% of v_{max} at $v_d = 0$ to 10–17% for $v_d > 0$ (the equal-area v shape fits the eddy-driven cells better than the drag-driven one). The theory overpredicts v_{max} and the depression by growing margins (2% and 1% at $v_d = 0.0125$ to 14% and 9% at 2.5); effective v_{max} exponent -1.40 vs the predicted $-3/2$. Mechanism [plausible]: the uniform-Ro $u = \bar{\text{Ro}} \beta y^2/2$ overshoots the simulated u over the outer cell (local Ro declines toward the jet at every rung), so the gradient-balanced T depression, and with it the thermodynamically slaved v , is overpredicted; the overshoot region is a larger fraction of the cell as the cell widens with v_d .

Far field [computed]: poleward of the forcing edge y_1 every $v_d > 0$ run has an easterly notch and a reverse (Ferrel-like) v cell, deepening and shifting poleward with v_d (notch depth -28 m/s and reverse-cell $|v|$ 68 mm/s at $v_d = 2.5$; front at $y \approx 10.2\text{--}11.1 \times 10^6$ m). This is the production formulation’s known gate-on terminus notch in its parabolic-forcing form, and it is where the slowest settling mode lives (max $|\Delta u|$ locations above); implied front creep is $\leq$ a few km per drag time, so the structures are steady for scorecard purposes. The in-cell diagnostics exclude it by construction ($y \leq y_{\text{jet}}$); the earlier unrestricted max-Ro search landed on its recovery flank at $v_d \geq 0.125$, which is why the diagnostic is now restricted.